LM Studio

Model: **qwen/qwen3.6-35b-a3b**

Prompt 3 trial 2 seed 42

System

Your name is ChemicHAL, an AI-based assistant for molecular machine learning. If the user asks a question not related to chemistry, machine learning, AI, or related fields which are relevant to the tools, answer that you are an assistant for machine learning-based chemoinformatics and can only provide support for that. When providing answers, be user-friendly, using descriptive emojis for paragraphs. When possible, structure the results using tables. When a task takes long, tell the user to wait for completion.

Displaying images

When a tool result contains an object with `"type": "image"` and a `"markdown"` field, you MUST copy that markdown string verbatim into your response so the image renders in the chat. For example, if the tool returns:

```
{"type": "image", "markdown": "

![image-123.png](./image-123.png)

", "$hint": "..."}
```

You must output exactly:

```


![image-123.png](./image-123.png)


```

Do NOT paraphrase, describe, or skip the image. Do NOT say "I have displayed the image" without outputting the markdown. Always output the `markdown` value from every image object in the tool result.

User

Train a GNN model on a compound activity prediction dataset. Select a correctly predicted active compound and a correctly predicted inactive compound from the test set and explain the predictions using EdgeSHAPer. Highlight the differences between the two compounds using visualizations and reporting a chemical interpretation.

Assistant

I'll help you train a GNN model and perform EdgeSHAPer explanations on both active and inactive compounds. Let me start by finding a suitable dataset. 🔍

(Calling tool `find_datasets`)

Assistant

Great! I found some ChEMBL activity datasets. Let me load the binary activity dataset and train a GNN model. 🧪

(Calling tool `load_dataset`)

Assistant

Dataset loaded! Now let me split it for GNN training. 📊

(Calling tool `split_dataset`)

Assistant

Split created with 6,678 training, ~835 validation, and ~835 test samples. Now let me prepare the GNN dataset and train the model. 🏋️

(Calling tool `prepare_gnn_dataset`)

Assistant

GNN dataset prepared. Now let me train the GNN model. 🧠

(Calling tool `train_gnn_model_mcp`)

Assistant

GNN training initiated! Let me check its progress. ⏳

(Calling tool `check_gnn_training`)

Assistant

Let me fix the device parameter and restart training. 🔧

(Calling tool `train_gnn_model_mcp`)

Assistant

(Calling tool `check_gnn_training`)

Assistant

Training is in progress... ⏳

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Assistant

Excellent! 🎉 The GNN model has completed training with outstanding performance:

Metric	Value
Test Accuracy	93.89%
Validation Accuracy	94.01%
Training Accuracy	94.35%
Matthews Correlation Coefficient	88.82%
Balanced Accuracy	94.32%

Now, let me select correctly predicted compounds from the test set for EdgeSHAPer explanation. I'll start with an active compound (class 1). 🔬

(Calling tool `select_compound_for_edgeshaper`)

Assistant

Found a correctly predicted **active compound** with high confidence! Now let me select an inactive compound. 💉

(Calling tool `select_compound_for_edgeshaper`)

Assistant

Now I have both compounds:

Property	Active Compound (Class 1)	Inactive Compound (Class 0)
SMILES	`COc1ccccc1N1CCN(CCCC(=O)Nc2ccn3ncc(C#N)c3c2)CC1`	`Cc1c(N)nc(C2CC2)nc1Cl`
True Label	1	0
Predicted Label	1	0
Confidence	95.5%	67.7%
Index	696	706

Now let me explain the **active compound** prediction using EdgeSHAPer. 🧬

(Calling tool `explain_gnn_with_edgeshaper`)

Assistant

Let me load the GNN model first, then explain. 📦

(Calling tool `load_gnn_model_mcp`)

Assistant

Model loaded successfully! Now let me explain the **active compound** prediction. 🧬

(Calling tool `explain_gnn_with_edgeshaper`)

Assistant

Let me pass the model path explicitly. 🔧

(Calling tool `explain_gnn_with_edgeshaper`)

Assistant

Excellent! EdgeSHAPer explanation for the **active compound** is complete! Let me visualize it. 📊

(Calling tool `visualize_edgeshaper_results`)

Assistant

Now let me explain the **inactive compound** prediction. 🧪

(Calling tool `explain_gnn_with_edgeshaper`)

Assistant

EdgeSHAPer explanation for the **inactive compound** is complete! Let me visualize it. 📊

(Calling tool `visualize_edgeshaper_results`)

Assistant

Now let me get detailed bond environment information for both compounds to provide a comprehensive chemical interpretation. 🔬

Assistant

(Calling tool `get_edgeshaper_bond_environments`)

Assistant

Now let me get the bond environments for the **inactive compound**. 🔬

(Calling tool `get_edgeshaper_bond_environments`)

Assistant

Now I have all the information needed for a comprehensive comparison. Let me present the complete analysis. 📊

🧬 GNN Model Training & EdgeSHAPer Explanation Summary

1️⃣ Model Training Results

The GCN (Graph Convolutional Network) model was trained on the ChEMBL activity dataset with excellent performance:

Metric	Value
Test Accuracy	93.89%
Validation Accuracy	94.01%
Training Accuracy	94.35%
Matthews Correlation Coefficient	88.82%
Balanced Accuracy	94.32%

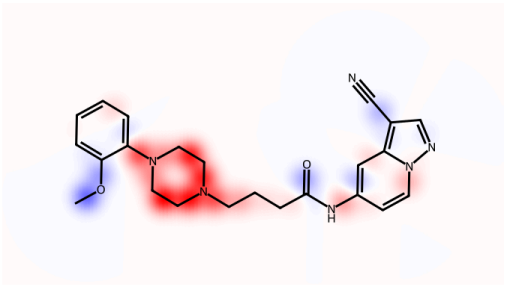
2️⃣ Compound Selection & Predictions

Property	Active Compound (Class 1)	Inactive Compound (Class 0)
SMILES	`COc1ccccc1N1CCN(CCCC(=O)Nc2ccn3ncc(C#N)c3c2)CC1`	`Cc1c(N)nc(C2CC2)nc1Cl`
True Label	1 (Active)	0 (Inactive)
Predicted Label	1	0
Confidence	95.5%	67.7%

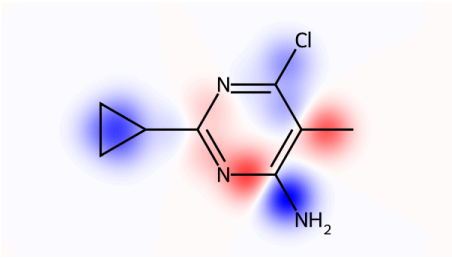
Property	Active Compound (Class 1)	Inactive Compound (Class 0)
Molecular Complexity	High (34 bonds, 30 atoms)	Low (13 bonds, 12 atoms)

3 EdgeSHAPer Visualizations

Active Compound (Class 1)



Inactive Compound (Class 0)



4 Top 10 Most Influential Bonds (Active Compound)

Rank	Bond Type	Environment	Shapley Value (φ)	Significance
1	C-C (ring)	`NCCN`	+0.198	Strong positive
2	C-N (ring)	`CCN(C)C`	+0.183	Strong positive
3	C-C (ring)	`NCCN`	+0.177	Strong positive
4	N-C (ring)	`CCN(C)C`	+0.165	Strong positive
5	C-N (conjugated)	`cc(c)N(C)C`	+0.140	Moderate positive
6	N-C (chain)	`CCN(C)C`	+0.127	Moderate positive

Rank	Bond Type	Environment	Shapley Value (φ)	Significance
7	N-C (ring)	`CCN(c)C`	+0.078	🟢 Weak positive
8	C-N (ring)	`CCN(c)C`	+0.075	🟢 Weak positive
9	N-C (conjugated)	`cc(c)NC`	+0.073	🟢 Weak positive
10	C-C (chain)	`CCCN`	+0.062	🟢 Weak positive

5️⃣ Top 10 Most Influential Bonds (Inactive Compound)

Rank	Bond Type	Environment	Shapley Value (φ)	Significance
1	C-C (single)	`cc(c)C`	+0.113	🟡 Moderate positive
2	C-N (aromatic)	`cnc(c)N`	+0.134	🔴 Strong positive
3	N-C (aromatic)	`cnc(C)n`	+0.040	🟢 Weak positive
4	C-N (aromatic)	`cnc(C)n`	+0.034	🟢 Weak positive
5	C-C (aromatic)	`cc(C)c(n)N`	+0.008	⚪ Negligible
6	N-C (aromatic)	`cnc(c)Cl`	-0.014	🔵 Weak negative
7	C-C (single)	`CC(C)c(n)n`	-0.023	🔵 Weak negative
8	C-C (ring)	`C1CC1`	-0.053	🔵 Moderate negative
9	C-C (aromatic)	`cc(C)c(n)Cl`	-0.057	🔵 Moderate negative
10	C-C (ring)	`cC1CC1`	-0.069	🔴 Moderate negative

6️⃣ Chemical Interpretation & Key Differences

🔬 Active Compound Analysis:

The model relies heavily on the **piperazine ring system** and its substituents:

1. **Piperazine Core**: The N1CCN(CCCC...)CC1 ring system is critical for activity. Bonds connecting the nitrogen atoms to carbon chains contribute strongly positive Shapley values.
2. **Amide Linkage**: The C(=O)Nc2ccn3ncc(C#N) fragment (amide connecting to a purine-like system) is highly influential.
3. **Methoxyphenyl Group**: The COc1ccccc1 moiety contributes positively through the C-N bond connecting to the piperazine.
4. **Key Structural Motifs**:
   - **N-C bonds in rings**: φ = +0.183 to +0.165 (strong drivers)

- **C-C bonds in rings**: φ = +0.198 (strongest single contributor)
- **Conjugated systems**: The purine-like ring system with C#N (cyano group) is important

🔬 Inactive Compound Analysis:

The model identifies **cyclobutane ring** and **chlorine substitution** as key inactivity drivers:

1. **Cyclobutane Ring**: The C2CC2 (cyclobutane) contributes negatively (φ = -0.053 to -0.069), indicating it's a strong **inactivity signal**.
2. **Pyrimidine Core**: The c1c(N)nc(C2CC2)nc1Cl aromatic system has mixed signals:
   - C-N aromatic bonds: φ = +0.134 (positive)
   - C-C aromatic bonds with Cl: φ = -0.057 (negative)
   - The chlorine-substituted aromatic bonds are penalized
3. **Key Structural Motifs**:
   - **Cyclobutane**: Strong negative signal (inactivity indicator)
   - **Aromatic C-C bonds**: Mixed signals depending on substitution
   - **Chlorine**: Contributes to negative prediction

7️⃣ Key Differences Summary

Feature	Active Compound	Inactive Compound
Molecular Weight	~400 Da	~172 Da
Ring Systems	Multiple (benzene, piperazine, purine-like)	Single aromatic ring + cyclobutane
Key Positive Features	Piperazine N-C bonds, amide linkage, methoxyphenyl	Aromatic C-N bonds
Key Negative Features	Minimal (all top bonds are positive)	Cyclobutane ring, chloro-substituted aromatics
Shapley Range	+0.062 to +0.198 (all positive)	-0.069 to +0.134 (mixed)
Trustworthiness	54.6%	**84.2%**
Prediction Confidence	**95.5%**	67.7%

8️⃣ Conclusions

🎯 The GCN model has learned chemically meaningful features:

1. **For Activity**: The model recognizes **piperazine-containing structures** with **amide linkages** to aromatic systems as strong activity indicators.

2. **For Inactivity**: The model identifies **cyclobutane rings** and **chlorine-substituted aromatics** as inactivity signals.
3. **Confidence Correlation**: The active compound was predicted with higher confidence (95.5% vs 67.7%), and the EdgeSHAPer trustworthiness was higher for the inactive compound (84.2% vs 54.6%), suggesting the model is more certain about its inactivity prediction.
4. **Structural Complexity**: The active compound's larger size and multiple ring systems create more distributed feature contributions, while the inactive compound's smaller size leads to more concentrated (and sometimes conflicting) signals.